```
plot_histogram(counts)
plt.show()
{'11': 508, '00': 516}
from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector

qc = QuantumCircuit(2)

# Create Bell state
qc.h(0)
qc.cx(0,1)

sv = Statevector.from_instruction(qc)
print(sv)
Statevector([0.70710678+0.j, 0.        +0.j, 0.        +0.j,
             0.70710678+0.j],
            dims=(2, 2))
```

Below is a **clean, minimal, concept-first explanation** of your code with **theory + intuition + debugging view**.

---

1. Quantum Measurement (Single Qubit)

What your circuit does

```python
qc = QuantumCircuit 1  1
qc.  0
qc.        0  0
```

Step-by-step (simple)

Step 1: Start

- Initial state = |0⟩

Step 2: Hadamard

```python
qc.  0
```

This creates:

$|0\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

👉 Meaning:

- 50% chance of 0
- 50% chance of 1
- But NOT decided yet (important)

Step 3: Measurement

```python
qc.        0  0
```

👉 Now the quantum state **collapses**

- You get either:
  - 0
  - 1

Output you got

```python
 '0'  490  '1'  534
```

👉 Interpretation:

- ~50% → 0
- ~50% → 1

✔️ Correct behavior

🔬 Theory (important for viva)

What is Quantum Measurement?

👉 Measurement = converting quantum → classical

Before measurement:

- System is in **superposition**

After measurement:

- System becomes **one definite value**

Key Rule

$$P(x) = |\alpha_x|^2$$

👉 Probability = square of amplitude

⚠️ Common Mistake

You might think:

> "State is both 0 and 1 physically"

❌ Wrong
✔️ It's a probability wave, not classical mixture

---

2. Entanglement (2 Qubits)

Now your second circuit:

```python
qc.  0
qc.   0  1
```

Step-by-step

Step 1: Hadamard on qubit 0

$$|00\rangle \to \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

👉 Only first qubit is in superposition

Step 2: CNOT

```python
qc.   0  1
```

👉 Rule:

- If control = 1 → flip target

So:

- |00⟩ → stays |00⟩
- |10⟩ → becomes |11⟩

Final state

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

👉 This is called a **Bell State**

Your Statevector

```python
0.707  0  0  0.707
```

👉 Meaning:

- Only |00⟩ and |11⟩ exist
- Equal probability

Measurement Output

```python
'00'  516  '11'  508
```

👉 Only two outcomes:

- 00
- 11

❗ Never:

- 01
- 10

---

3. What is Entanglement (core idea)

👉 Entanglement = **strong correlation**

If you measure:

- Qubit 0 = 0 → Qubit 1 = 0
- Qubit 0 = 1 → Qubit 1 = 1

Even if separated

Formal meaning

👉 System cannot be written as:

[redacted]

✔️ They are **linked**

🔬 Important Theory (Viva Ready)

Bell State

- Created using:
  - Hadamard
  - CNOT

Used in:

- Quantum Teleportation
- Superdense Coding

---

4. Measurement vs Entanglement

Concept	Meaning
Measurement	Collapse state
Entanglement	Correlated qubits

---

5. Key Insight (VERY IMPORTANT)

👉 Superposition = multiple possibilities
👉 Entanglement = linked outcomes

---

6. Debugging Understanding

If results look wrong:

For measurement:

- Did you apply H? → should be 50-50
- No H → always 0

For entanglement:

- Missing CNOT → no correlation
- Wrong control-target → wrong pairs

---

7. One-line intuition

- Measurement → "pick one result"
- Entanglement → "results are connected"

If you want deeper clarity

I can:

- ↳ show full amplitude evolution table
- ↳ compare separable vs entangled mathematically
- or ↳ give viva Q&A format

Just specify.